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worked_example_z_audit <- sdf_sql(
  sc,
  "
  WITH selected_parties AS (

    SELECT
      party_id,
      observation_date,
      label_3m,

      /* FINAL DATABASE VALUES */

      active_accounts_z_model,
      active_accounts_z_available_flag,

      avg_utilization_z_model,
      avg_utilization_z_available_flag,

      weighted_utilization_z_model,
      weighted_utilization_z_available_flag,

      avg_utilization_change_model,
      weighted_utilization_change_model,

      utilization_change_available_flag,
      utilization_recent_months_available,
      utilization_prior_months_available

    FROM bu_frd.model_features_ab_final_2022_2025

    WHERE party_id IN (
      <PARTY_ID_LABEL_0>,
      <PARTY_ID_LABEL_1>
    )
  ),

  /* =====
  ACTIVE ACCOUNTS
  ===== */

  active_calc AS (

    SELECT
      b.party_id,
      b.observation_date,

      AVG(
        CASE

```

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        WHEN b.month_offset BETWEEN 3 AND 11
        THEN b.active_accounts
    END
) AS active_prior_avg,

STDDEV_SAMP(
    CASE
        WHEN b.month_offset BETWEEN 3 AND 11
        THEN b.active_accounts
    END
) AS active_prior_std,

AVG(
    CASE
        WHEN b.month_offset BETWEEN 0 AND 2
        THEN b.active_accounts
    END
) AS active_recent_avg

FROM bu_frd.model_party_behavior_monthly_2022_2025 b

INNER JOIN selected_parties s
    ON b.party_id = s.party_id
    AND b.observation_date = s.observation_date

GROUP BY
    b.party_id,
    b.observation_date
),

/* =====
AVG + WEIGHTED UTILIZATION
===== */

util_calc AS (

SELECT
    u.party_id,
    u.observation_date,

    /* ----- AVG UTILIZATION ----- */

    COUNT(
        CASE
            WHEN u.month_offset BETWEEN 3 AND 11
            AND u.avg_utilization IS NOT NULL
            THEN 1
        END
    )

```

```

) AS avg_util_prior_n,

COUNT(
  CASE
    WHEN u.month_offset BETWEEN 0 AND 2
      AND u.avg_utilization IS NOT NULL
    THEN 1
  END
) AS avg_util_recent_n,

AVG(
  CASE
    WHEN u.month_offset BETWEEN 3 AND 11
    THEN u.avg_utilization
  END
) AS avg_util_prior_avg,

STDDEV_SAMP(
  CASE
    WHEN u.month_offset BETWEEN 3 AND 11
    THEN u.avg_utilization
  END
) AS avg_util_prior_std,

AVG(
  CASE
    WHEN u.month_offset BETWEEN 0 AND 2
    THEN u.avg_utilization
  END
) AS avg_util_recent_avg,

/* ----- WEIGHTED UTILIZATION ----- */

COUNT(
  CASE
    WHEN u.month_offset BETWEEN 3 AND 11
      AND u.weighted_utilization IS NOT NULL
    THEN 1
  END
) AS weighted_util_prior_n,

COUNT(
  CASE
    WHEN u.month_offset BETWEEN 0 AND 2
      AND u.weighted_utilization IS NOT NULL
    THEN 1
  END

```

```

) AS weighted_util_recent_n,

AVG(
  CASE
    WHEN u.month_offset BETWEEN 3 AND 11
    THEN u.weighted_utilization
  END
) AS weighted_util_prior_avg,

STDDEV_SAMP(
  CASE
    WHEN u.month_offset BETWEEN 3 AND 11
    THEN u.weighted_utilization
  END
) AS weighted_util_prior_std,

AVG(
  CASE
    WHEN u.month_offset BETWEEN 0 AND 2
    THEN u.weighted_utilization
  END
) AS weighted_util_recent_avg

FROM bu_frd.model_party_utilization_monthly_2022_2025 u

INNER JOIN selected_parties s
  ON u.party_id = s.party_id
  AND u.observation_date = s.observation_date

GROUP BY
  u.party_id,
  u.observation_date
),

calculated AS (

SELECT
  s.party_id,
  s.observation_date,
  s.label_3m,

  /* =====
  ACTIVE ACCOUNTS
  ===== */

  a.active_prior_avg,
  a.active_prior_std,
  a.active_recent_avg,

```

```

a.active_recent_avg
- a.active_prior_avg
AS active_change_recalc,

CASE
  WHEN a.active_prior_std > 0
  THEN
    (
      a.active_recent_avg
      - a.active_prior_avg
    )
    /
    (
      a.active_prior_std / SQRT(3.0)
    )
END AS active_z_raw_recalc,

```

```

/* =====
AVG UTILIZATION
===== */

```

```

u.avg_util_prior_n,
u.avg_util_recent_n,

```

```

u.avg_util_prior_avg,
u.avg_util_prior_std,
u.avg_util_recent_avg,

```

```

u.avg_util_recent_avg
- u.avg_util_prior_avg
AS avg_util_change_recalc,

```

```

CASE
  WHEN u.avg_util_recent_n = 3
  AND u.avg_util_prior_n >= 6
  AND u.avg_util_prior_std > 0
  THEN
    (
      u.avg_util_recent_avg
      - u.avg_util_prior_avg
    )
    /
    (
      u.avg_util_prior_std / SQRT(3.0)
    )
END AS avg_util_z_raw_recalc,

```

```

/* =====
WEIGHTED UTILIZATION
===== */

```

```

u.weighted_util_prior_n,
u.weighted_util_recent_n,

u.weighted_util_prior_avg,
u.weighted_util_prior_std,
u.weighted_util_recent_avg,

u.weighted_util_recent_avg
  - u.weighted_util_prior_avg
  AS weighted_util_change_recalc,

```

```

CASE
  WHEN u.weighted_util_recent_n = 3
    AND u.weighted_util_prior_n >= 6
    AND u.weighted_util_prior_std > 0
  THEN
    (
      u.weighted_util_recent_avg
      - u.weighted_util_prior_avg
    )
    /
    (
      u.weighted_util_prior_std / SQRT(3.0)
    )
  END AS weighted_util_z_raw_recalc,

```

```

/* =====
VALUES STORED IN FINAL DATABASE
===== */

```

```

s.active_accounts_z_model,
s.active_accounts_z_available_flag,

s.avg_utilization_change_model,
s.avg_utilization_z_model,
s.avg_utilization_z_available_flag,

s.weighted_utilization_change_model,
s.weighted_utilization_z_model,
s.weighted_utilization_z_available_flag,

```

```

        s.utilization_change_available_flag,
        s.utilization_recent_months_available,
        s.utilization_prior_months_available

FROM selected_parties s

LEFT JOIN active_calc a
  ON s.party_id = a.party_id
  AND s.observation_date = a.observation_date

LEFT JOIN util_calc u
  ON s.party_id = u.party_id
  AND s.observation_date = u.observation_date
)

SELECT
  *,

  /* =====
  EXPECTED FINAL Z AFTER FROZEN WINSORIZATION CAPS
  ===== */

  CASE
    WHEN active_z_raw_recalc IS NULL
      THEN NULL
    WHEN active_z_raw_recalc < -13.27906
      THEN -13.27906
    WHEN active_z_raw_recalc > 15.01716
      THEN 15.01716
    ELSE active_z_raw_recalc
  END AS active_z_after_cap_recalc,

  CASE
    WHEN avg_util_z_raw_recalc IS NULL
      THEN NULL
    WHEN avg_util_z_raw_recalc < -29.78358
      THEN -29.78358
    WHEN avg_util_z_raw_recalc > 33.76045
      THEN 33.76045
    ELSE avg_util_z_raw_recalc
  END AS avg_util_z_after_cap_recalc,

  CASE
    WHEN weighted_util_z_raw_recalc IS NULL
      THEN NULL
    WHEN weighted_util_z_raw_recalc < -37.89802
      THEN -37.89802
    WHEN weighted_util_z_raw_recalc > 40.84790

```

```
        THEN 40.84790
        ELSE weighted_util_z_raw_recalc
    END AS weighted_util_z_after_cap_recalc
```

```
FROM calculated
```

```
ORDER BY label_3m, party_id
"
```

```
) %>%
```

```
collect()
```

```
worked_example_z_audit
```